
SE LAB 3

NAME : RISHI J MULE

SRN : PES1UG24AM224

SECTION : D

Faculty Research Grant & Publication Tracker

Architecture Selection

We chose **Layered Architecture** for the Faculty Research Grant & Publication Tracker.

Reason 1 – Separation of Responsibilities

Layered Architecture provides a clear separation between presentation, business logic, and data management. This is suitable for the system because grant management, publication tracking, approval workflows, and analytics can be handled as separate components while maintaining a structured system.

Reason 2 – Maintainability

The separation of responsibilities makes the system easier to maintain and modify. Changes to grant management or publication management can be made within their respective components without unnecessarily affecting other parts of the system.

Security Advantage

Sensitive research and financial information can be protected within the business and data layers. Access to grant information, financial approvals, and audit records can be controlled before requests reach the underlying data. This is important because the system must maintain a tamper-evident audit ledger for financial approvals and journal-status modifications.

Performance Benefit

Separating presentation, business processing, and data management allows requests to be handled by the appropriate layer. This can improve the efficiency of frequently accessed operations such as retrieving grant balances, expense information, and publication records.